

Session 4: T-Route Configuration and Open-Loop Simulation

Configure and run T-Route in standalone mode using the two routing schemes: Muskingum-Cunge (MC) and Diffusive Wave (DW). The MC configuration (`troute.yaml`) is provided in the workshop repository and is used as the baseline; the DW configuration reuses the same file with a small `hybrid_parameters` block added. All paths below assume the Docker workflow layout used in Sessions 2 and 3.

Muskingum-Cunge (MC) Open-Loop Run

Use the MC `troute.yaml` from the workshop repository as-is. Key parameters already configured:

- **Network:** `/workspace/data/Coralville2/config/Coralville2_subset.gpkg`
- **Lateral inflow (qlat):** `/workspace/data/Coralville2/outputs/ngen/` (cat-* files from the NextGen run in Session 3)
- **Simulation start:** `2020-01-01 00:00:00, dt = 300 s, nts = 105408`
- **Output:** `/workspace/data/Openloop/MC/Output/` as a single `.nc` file at 60-minute internal frequency
- **Data assimilation: disabled** (open-loop)

Run T-Route standalone (MC)

From the directory containing `troute.yaml`, launch the routing engine:

```
python -m nwm_routing -f -V4 troute.yaml
```

Flag reference for this command:

- `-f` : run in file mode (read inputs from disk).
- `-V4` : use the V4 configuration schema (matches the YAML structure shown above).
- `troute.yaml` : path to the configuration file.

Diffusive Wave (DW) Open-Loop Run

The DW run reuses the MC `troute.yaml` with one addition: a `hybrid_parameters` block that enables diffusive routing on the mainstem and points at the diffusive domain and topobathy files.

Step 1: Add the `hybrid_parameters` block

Insert the block below into `troute.yaml` immediately after the `supernetwork_parameters` section (i.e., right after the mainstem: "mainstem" column entry). Update the `stream_output_directory` at the bottom of the file to a DW-specific path (e.g., `/workspace/data/Openloop/DW/Output/`) so MC and DW results are kept separate.

```
mainstem: "mainstem"

hybrid_parameters:
  run_hybrid_routing: True
  diffusive_domain : /workspace/data/Openloop/DW/Domain/Domain.yaml
  use_natl_xsections: True
  topobathy_domain  : /workspace/data/Openloop/DW/Domain/Mainstem.parquet
```

Parameter reference for the added block:

- `run_hybrid_routing`: True : enable hybrid MC/DW routing (DW on the mainstem, MC elsewhere).
- `diffusive_domain`: YAML file listing the mainstem reaches to be routed with the diffusive solver.
- `use_natl_xsections`: True : use natural cross-sections from the topobathy dataset (rather than trapezoidal approximations).
- `topobathy_domain`: parquet file containing surveyed cross-section geometry for the mainstem.

Step 2: Run T-Route standalone (DW)

With the modified `troute.yaml` in place, launch the same command as for the MC run:

```
python -m nwm_routing -f -V4 troute.yaml
```